

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**End Semester Examination – Summer 2019****Course: B. Tech Electrical****Sem: III****Subject Name: Measurement and Instrumentation****Subject Code: BTEEC304****Max Marks: 60****Date: 30/05/2019****Duration: 3 Hr.****Instructions to the Students:**

1. Solve **ANY FIVE** questions out of the following.
2. The level question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What is error in measurement? Write in brief types of errors.	CO1	
B) Explain types of various standards used in the electrical measurement.	CO1	
C) Give classification of various analog measuring instruments.	CO1	
		12
Q.2 Solve the following.		
A) Write about instrument transformers & their application in the extension of instrument range.	CO1	
B) Write a note on electrodynamicometer type single phase wattmeter.	CO2	
Q. 3 Solve the following.		12
A) State different methods of measuring low, medium and high resistances. Explain Meggar method in detail.	CO2	
B) Derive the balancing condition for Maxwell inductance bridge with neat diagram of bridge.	CO2	
Q.4 Solve Any Two of the following.		12
A) With neat block diagram explain Ramp type digital voltmeter.	CO2	
B) Write a note on electronic multimeter.	CO2	
C) What is the use of power analyzer? Explain its working.	CO2	
Q. 5 Solve the following.		12
A) Explain the transducer used for measurement of linear displacement.	CO3	
B) Write a note on methods of data transmission.	CO3	

Q. 6 Solve the following.

12

- A) Explain in detail digital storage oscilloscope with its applications.
- B) Write a short note on X-Y recorders.

CO3

CO3

***** End *****